Hasib Zunair

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Work Experience

Pave Toronto, Canada
Senior Machine Learning Engineer *May 2025 – Present*

- **Delivered >80% automation** for exterior vehicle inspection by designing multimodal fusion and spatiotemporal damage detection system; achieving 92% precision, 87% recall and reducing turnaround time by 85% compared to human inspectors, enabling large-scale fleet inspections and repair cost estimation without manual intervention.
- Architected neural network pipelines for automating component identification workflow with 89% accuracy and image quality analysis reducing 85% compute cost via distilling VLMs and capability for offline on-device usage.
- Implemented multi-task learning to predict granular damage attributes with a unified model; achieving 90% precision reducing manual inspection steps and improving structured metadata for vehicle condition reports.

Decathlon Montreal, Canada
Machine Learning Engineer *Sept. 2020 – June 2024*

- Built real-time basketball video object tracking pipeline; optimized with quantization, pruning, and distillation, cutting memory + latency by 30%; designed model-assisted labeling workflow, reducing annotation cost by 85%.
- Implemented semi-supervised learning using EfficientNets to train on unlabeled data, resulting 10% accuracy, 5× lower cloud usage; developed generative AI models using GANs, pix2pixHD, OpenPose, improving realism by 20% and fine-tuned YOLO for bike brand detection from marathon images, driving faster, cheaper market insights.
- **Tools**: Python, PyTorch, OpenCV, ONNX, CoreML, FastAPI, Docker, GCP.

Concordia University Montreal, Canada
Graduate Machine Learning Researcher *Sept. 2019 – Dec. 2024*

- Led publications and presented at top-tier conferences (WACV, BMVC), oral presentation at BMVC'22 (top 5% acceptance rate), journals (IEEE TMI, IF: 10.6) and workshops (CVPR, ICML), resulting in 1600+ citations.
- Designed and implemented novel 2D/3D deep learning algorithms, outperforming state-of-the-art in accuracy, data efficiency, and compute; tailored vision models like ConvNet, ViT, DINO, YOLO, CycleGAN using unsupervised, self-supervised, and zero-shot learning for image recognition, generation, detection, and segmentation.
- Mentored 15 students from undergraduate to Ph.D. level through research projects, publishing at several journals.

Ericsson Montreal, Canada
Machine Learning Specialist *Oct. 2021 – Mar. 2022 & Feb. 2024 – June 2024*

- Led hands-on training for GPT-style LLMs, fine-tuning and data science tooling; mentored via pair programming, code reviews, and experiment design to drive successful end-to-end project delivery.
- Guided 21 Ericsson staff in building seq2seq models (Transformers, LSTM autoencoders) for time-series forecasting and anomaly detection using PyCaret, PyOD and Darts, improving model performance and production readiness.

Skills

- **Programming Languages**: Python, C++.
- **Tools & Libraries**: PyTorch, Unsloth, TensorFlow/Keras, scikit-learn, OpenCV, ONNX, CoreML.
- **Cloud Infrastructure and MLOps**: GCP, FastAPI, Docker, Gradio, GitHub Actions, Kubernetes, RunPod.

Projects & Publications & Open-Source Contributions

- **Multi-Modal AI (2025)**: Created domain-specific datasets to enable two systems: Afiyah, a multimodal OCR pipeline using LLM fine-tuning and RAG-style retrieval to extract and explain harmful ingredients. FitLensAI, a Llama 3-based vision-language model fine-tuned with LoRA for multilingual, multi-turn fitness conversations from workout images.
- **Unsupervised Object Localization at BMVC (2024)**: Created PEEKABOO using DINO to segment unfamiliar objects without any additional training, that is accurate, faster and lighter than competing state-of-the-art methods.
- **AICITY competition at CVPR (2022)**: Led a team for VISTA, by training Vision Transformers (ViTs) on synthetic data, to recognize and count products in videos for retail checkout to improve operational efficiency, achieving 3rd place.
- **Contributions to core ML/CV libraries with >75K GitHub stars**: Added Python code to Kornia (MS-SSIM loss), TensorFlow (3D image classification), and YOLOv6 (ONNX export fix), driving higher accessibility and usability.

Education

- **Concordia University** Doctor of Philosophy (Ph.D.) and M.A.Sc in Machine Learning & Artificial Intelligence
- **North South University** B.Sc. in Electrical & Computer Engineering